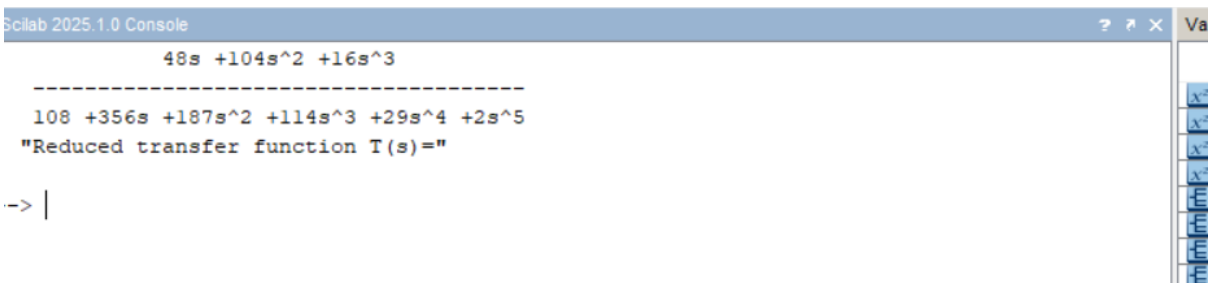
Program code:

(i) The block diagram reduction technique

```
clc;
s=poly(0,'s');
n1=[2*s];
d1=[3+s^2];
h1=syslin('c',n1/d1);
n2=[2];
d2=[s+4];
h2=syslin('c',n2/d2);
n3=[4];
d3=[s+6];
h3=syslin('c',n3/d3);
n4=[s];
d4=[1+2*s];
h4=syslin('c',n4/d4);
h5=h3/.h4; //h3 and h4 in negative feedback loop
h6=h1*h5; //h1 and h5 are in series
h7=h6/.h3; //h6 and h3 in negative feedback loop
disp(h7,"Reduced transfer function T(s)=");
```

OUTPUT:

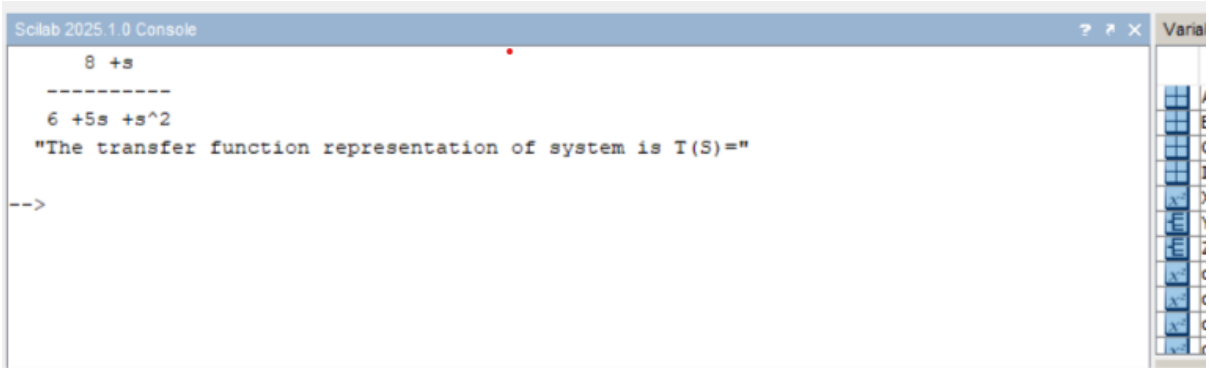


(ii) The state model reduction technique

a) for single input and single output (SISO) system

```
clc;
s=poly(0,'s');
A=[0 1;-6 -5];
B=[0;1];
C=[8 1];
[n1,n2]=size(A);
I=eye(n1,n2); //Identity matrix
X=s*I-A;
phi=inv(X); //Inverse of Matrix
Y=C*phi;
Z=Y*B;
//sys=tf2ss(Z)
disp(Z,"The transfer function representation of system is T(S)=");
```

OUTPUT:



b) for multiple input and multiple output (MIMO) system

```
clc;
clear;
close;
A=[0 1;-25 -4];
B=[1 1;0 1];
C=[1 0;0 1];
D=[0 0;0 0];
```

```
H=syslin('c',A,B,C,D);
disp(H,'The given state model matrices are');
disp(A,'A=');
disp(B,'B=');
disp(C,'C=');
disp(D,'D=');
G=clean(ss2tf(H));
disp(G,'The obtained transfer functions are');
```

Simulated output:

